Quadratic Programming

The general QP:

$$\min\ q(x) = \frac{1}{2}x^T G x + x^T c$$
$$\text{s.t.}\quad A_e x = b_e$$
$$A x \geq b$$
$$x \in \mathbb{R}^n$$

equivalently:

$$\min\ q(x) = \frac{1}{2}x^T G x + x^T c$$
$$\text{s.t.}\quad a_i^T x = b_i,\ i \in \mathcal{E}$$
$$a_i^T x \geq b_i,\ i \in \mathcal{I}$$
$$x \in \mathbb{R}^n$$

Of particular interest to us: $G \geq 0$

First consider the equality constrained problem.

$$\min\ q(x) = \frac{1}{2}x^T G x + x^T c \qquad L = \frac{1}{2}x^T G x + x^T c - \lambda(Ax - b)$$
$$\text{s.t.}\quad Ax = b$$
$$x \in \mathbb{R}^n$$

The KKT conditions are $Gx + c - A^T\lambda = 0$ & $Ax = b$ which can be written:

$$\begin{bmatrix} G & -A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ \lambda^* \end{bmatrix} = \begin{bmatrix} -c \\ b \end{bmatrix}$$

Our assumptions:

(1) $G \geq 0$

(2) A is full rank

(3) $Z^T G Z > 0$ where Z is the matrix whose columns form a basis for the null space of A.

(3) Every feasible direction is a direction of positive curvature (of q).

Example $\min\ q = x^2 + y^2$

s.t. $3x + y = 3$

$$q(x,y) = x^2+y^2 = \frac{1}{2}[x\ y]\overset{G}{\begin{bmatrix}2&0\\0&2\end{bmatrix}}\begin{bmatrix}x\\y\end{bmatrix} + [x\ y]\overset{c}{\begin{bmatrix}0\\0\end{bmatrix}}$$

$$3x+y = \overset{A}{[3\ 1]}\begin{bmatrix}x\\y\end{bmatrix} = \overset{b}{3}$$

KKT:
$$\begin{bmatrix}2&0&-3\\0&2&-1\\3&1&0\end{bmatrix}\begin{bmatrix}x^*\\y^*\\\lambda^*\end{bmatrix} = \begin{bmatrix}0\\0\\3\end{bmatrix} \Rightarrow \begin{matrix}x^* = 0.9\\y^* = 0.3\\\lambda^* = 0.6\end{matrix}$$

A computational variant:

Suppose x is not necessarily optimal. Find a step p^* so that $x^* = x + p^*$. Apply the KKT conditions to x^*. We get:

$$\begin{bmatrix}G&A^T\\A&0\end{bmatrix}\begin{bmatrix}-p^*\\\lambda^*\end{bmatrix} = \begin{bmatrix}Gx+c\\Ax-b\end{bmatrix}$$

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suppose $x = \begin{bmatrix}1\\1\end{bmatrix}$, $Gx+c = \begin{bmatrix}2\\2\end{bmatrix}$, $Ax-b = 1$

$$\begin{bmatrix}2&0&3\\0&2&1\\3&1&0\end{bmatrix}\begin{bmatrix}p_1^*\\p_2^*\\\lambda^*\end{bmatrix} = \begin{bmatrix}2\\2\\1\end{bmatrix} \Rightarrow \begin{matrix}p_1^* = -0.1 & p_2^* = -0.7\\ x^* = x + p^* = \begin{bmatrix}0.9\\0.3\end{bmatrix}\end{matrix}$$

Lemma. Let Z be the matrix whose columns form a basis for the null space of full rank matrix A. Suppose $Z^T G Z > 0$, then

$$K = \begin{bmatrix} G & A^T \\ A & 0 \end{bmatrix} \text{ is invertible.}$$

Proof. Suppose w and v satisfying $K\begin{bmatrix} w \\ v \end{bmatrix} = 0$, we show that $w=0$, $v=0$ is the unique solution. (therefore K is invertible).

We have $Gw + A^T v = 0$ and $Aw = 0$. $*$

$$0 = [w^T \ v^T]\begin{bmatrix} G & A^T \\ A & 0 \end{bmatrix}\begin{bmatrix} w \\ v \end{bmatrix} = [w^T \ v^T]\begin{bmatrix} Gw + A^T v \\ Aw \end{bmatrix} = w^T G w$$

But $w \in \text{Null}(A)$ so $\exists\, u \ni w = Zu$.

$$0 = w^T G w = (Zu)^T G (Zu) = u^T Z^T G Z u$$

As $Z^T G Z > 0 \Rightarrow u = 0$ and $w = Zu = 0$.

Finally $Gw + A^T v = 0 \Rightarrow v = 0$. $\square$

Theorem. Let A have full row rank and assume $Z^T G Z > 0$. Then x^* satisfying

$$\begin{bmatrix} G & A^T \\ A & 0 \end{bmatrix}\begin{bmatrix} x^* \\ \lambda^* \end{bmatrix} = \begin{bmatrix} c \\ -b \end{bmatrix}$$

is the unique global min of ECQP.

The projected Gradient Method for solving KKT system

Given: x satisfying $Ax=b$

: projection operator P (onto NULL A)

Compute: $r = Gx+c$, $g = Pr$, $d = -g$

Repeat until $r^T g < tol$

$\alpha \leftarrow r^T g / d^T G d$

$x \leftarrow x + \alpha d$

$r^+ \leftarrow r + \alpha G d$

$g^+ \leftarrow P r^+$

$\beta \leftarrow (r^+)^T g / r^T g$

$d \leftarrow -g^+ + \beta d$

$g \leftarrow g^+$

$r \leftarrow r^+$

$P = Z(Z^T Z)^{-1} Z^T$

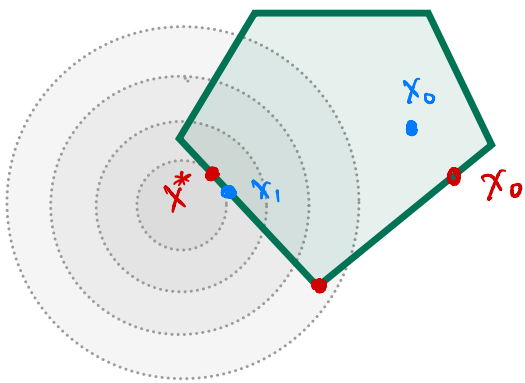
$P = I - A^T (AA^T)^{-1} A$

$Pr = r - A^T \underbrace{(AA^T)^{-1} A r}_{v}$

$$\boxed{\begin{aligned} Pr &= r - A^T v \\ (AA^T) v &= Ar \end{aligned}}$$

Active Set Method

If we know the active constraints at x^* then we simply solve $\{a_i^T x = b_i\}$ for $i \in A(x^*)$.

$\Rightarrow$ A search for the active set at x^*.

Concept: Intelligently search combinations of active constraints (the working set), improving $g(x)$ with finite feasible sequence $x_k \to x^*$.